

# Day 14: Expectation, Variance and Binomial Distribution

## Understanding Average Behavior and Uncertainty

100 Days of Data Science

### Why These Concepts Matter

In Data Science and Machine Learning, we are not interested in one random outcome. We care about:

- Average behavior
- Risk and uncertainty
- Probability of success or failure

Expectation and variance describe average and spread, while binomial distribution models yes/no outcomes.

### Expectation (Expected Value)

Expectation tells the long-term average value of a random variable.

It answers: **What value should we expect on average after many trials?**

### Formula

For discrete random variable  $X$ :

$$E(X) = \sum x \cdot P(x)$$

## Dice Example

Rolling a dice:

$$E(X) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Meaning: After many rolls, average result approaches 3.5.

## Real-Life Example

A shop gets:

- 40 customers some days
- 60 customers some days

Expected customers 50 per day.

Businesses plan resources using expectation.

## Expectation in Machine Learning

ML models try to:

- Minimize expected error
- Predict expected outcome

Loss functions measure expected prediction mistakes.

## Variance

Variance measures how much values differ from the average.

It answers: **How spread out is the data?**

## Formula

$$Var(X) = E[(X - \mu)^2]$$

Where  $\mu$  is expected value.

## Intuition

- Low variance  $\rightarrow$  stable outcomes
- High variance  $\rightarrow$  risky and unpredictable

## Real-Life Example

Two businesses both earn 50,000 average:

Business A: 48k–52k (low variance) Business B: 5k–95k (high variance)

Same average — but B is much riskier.

## Variance in Machine Learning

- High variance models overfit data
- Low variance models generalize better

This is the famous bias–variance tradeoff.

## What is Binomial Distribution?

Binomial distribution models number of successes in fixed number of trials where:

- Each trial has two outcomes (success/failure)
- Probability remains same
- Trials are independent

## Common Examples

- Coin toss (head or tail)
- Customer clicks ad or not
- Email is spam or not
- Product defective or good

## Binomial Formula

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Where:

- $n$  = number of trials
- $k$  = number of successes
- $p$  = probability of success

## Simple Example

If probability of success = 0.3 Trials = 10

We can calculate chance of exactly 3 successes.

## Real-Life Business Example

An online ad has 20% chance of click.

Out of 100 users: Binomial distribution predicts number of clicks.

Used in marketing performance analysis.

## Binomial in Machine Learning

Classification problems are binomial in nature:

- Spam / Not spam
- Fraud / Not fraud
- Disease / No disease

Logistic regression models binomial probabilities.

# Connection Between Expectation and Binomial

Expected successes in binomial:

$$E(X) = np$$

Example: 100 users  $\times$  0.2 click rate = 20 expected clicks.

## Why This Is Powerful

- Predict outcomes
- Measure risk
- Optimize ML decisions
- Plan business strategies

## Common Mistakes

- Confusing probability with certainty
- Ignoring variance
- Assuming averages tell full story

## Final Takeaway

Expectation predicts average behavior. Variance measures uncertainty. Binomial models success and failure.

Together they form the backbone of Machine Learning predictions.